

See It Again Before You Forget It.

Spaced Repetition · Ebbinghaus · Cepeda, Pashler & colleagues

"I knew everything on Sunday.
By Wednesday I remembered maybe half."

— That's not a memory problem. That's the forgetting curve doing exactly what it does.

In 1885, Hermann Ebbinghaus discovered that **memory decays exponentially after learning, following a predictable curve**. By one hour after studying, you've forgotten roughly 50% of new material. By one week, 75–80%. But every time you successfully retrieve information **just before forgetting it**, you reset the curve at a higher baseline — and the next decay is slower.

The Research

Cepeda, Pashler, Vul, Wixted, and Rohrer (2006) synthesized 254 studies and 14,000 participants. Their finding: distributed practice consistently outperforms massed practice (cramming) for long-term retention, with the optimal gap between sessions scaling with how far in the future you need to recall the material.

254 studies in Cepeda et al. (2006) — the largest meta-analysis of spacing research ever assembled	10–30% retention gain from spaced vs. massed practice on delayed tests (Cepeda et al., 2006)	1→3→7 days — basic spacing sequence for a one-week exam. Three encounters bracketing the decay window.
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For an exam one week away: **study today → review in 1 day → review in 3 more days → exam**. Three encounters, each preventing full decay, each rebuilding the memory trace stronger. Cramming the night before is one encounter with a week of decay in between.

The spacing sweet spot: Review when you've forgotten roughly 20–30% of the material — not when it's fresh (too easy, no encoding benefit) and not when it's completely gone (mostly re-learning). The slight effort of partial recall is the signal you're in the right window.

The Protocol That Works

When	What to Do & Why
Day 1 (study session)	Initial encoding. Engage actively — ask questions, generate examples. Don't try to memorize everything; just expose yourself with genuine attention.

When	What to Do & Why
Day 2 (24 hours later)	First spacing event — right at the first major decay point. Brief retrieval practice session (not re-reading). Covers what Day 1 encoded before it fades.
Day 4–5	Second spacing event. Practice quiz format without notes. The effort of partial recall is the mechanism — struggle signals learning.
Day before exam	Final spacing event — consolidation only. If prior sessions went well, this should feel mostly like recognition, not re-learning.

Spread sessions, not just study time: Five 30-minute sessions across a week produce more retention than one 150-minute session — even with identical total time. The spacing between sessions is the variable that matters.

Spaced repetition + retrieval practice: The two most evidence-backed techniques in cognitive psychology are most powerful when combined. Each spacing event should be a retrieval attempt — not re-reading. Cover your notes, recall, check, repeat.

The cramming trap: Cramming works for next-day recall. It fails for any professional skill requiring the knowledge later. A 30-minute review three days before an exam outperforms three hours of cramming the night before on any delayed test.

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